

LABLAB.AI × ALPACA AI TRADING AGENTS HACKATHON · TEAM GATED AGENT

Gated Agent

An options agent whose every order must survive its own red team.

Public repo `Theodore-Liu/gated-agent` · live demo `gated-agent-live.streamlit.app`

Competition account `PA32VHB05A0B` — every trade a defined-risk option spread.

The inversion

Most trading-agent demos put the LLM in charge of finding alpha.

We put it in charge of saying no.

- Signal: deliberately a textbook toy — Faber 10-month SMA on SPY / QQQ / IWM / AAPL / NVDA. Public literature, four fields.
- Deterministic code maps signal → defined-risk spread (Δ -targeted legs).
- The LLM's only power is **veto**. It cannot size, price, or place anything.

Arithmetic that cannot be talked out of

Six gates, before the LLM ever sees the order:

worst-case loss $\leq 1\%$ per trade

5% position cap

−2% daily halt

idempotent dedup

direction-flip guard (no hedged books)

liquidity floor

Exits are **pre-registered**: R1–R4 frozen in `config/close_rules.json` *before* the contest window — git history proves the rules predate the runs.

The red team

Before each order, an LLM wired **read-only** into Alpaca's official MCP server answers three questions from live account + market data:

- 1 What exactly is lost in the worst case — computed or guessed?
- 2 Does this stack dangerously with *this book's* existing positions?
- 3 What does the exit cost through this spread tomorrow?

Real rehearsal veto: per-order math fine at <1% max loss — vetoed anyway for stacking a third correlated spread sharing a strike with an open position. **The protocol recomputes the verdict from per-question verdicts: the model cannot approve past its own failed question.**

Every decision traceable

- Append-only JSONL ledger: signal → gates → red-team transcript → order intent → **broker receipt**, reconciled against the account every round.
- **Negative control**: a seeded random signal runs the identical pipeline, shadow-only, structurally unable to reach the order path.
- Torn-tail quarantine, crash-safe dedup, market-holiday awareness — it ran the contest week unattended.

WHAT WE ARE PROUDEST OF

We attacked it before the market could

19

real defects

Tests: **110** → **219**

Two adversarial reviews, pre-contest (docs/ADVERSARIAL-REVIEW.md) — a dedup gate that didn't dedup, a dead loss-halt, a close rule that wasn't unconditional, a negative control silently inheriting the live book — **each fixed with a regression test that fails on the pre-fix tree.**

*A trustworthy agent isn't one that never had bugs.
It's one that hunts its own.*

What we'd claim

- Discipline as the product: pre-registration, negative controls, veto-only LLM power, receipt-level traceability.
- Unattended live operation on the competition account from day 1, every order a filled multi-leg spread with its receipt in the book.
- Everything in this deck is verifiable in the public repo's git history.

Built with Alpaca Trading API + official MCP server + CLI (mLeg).